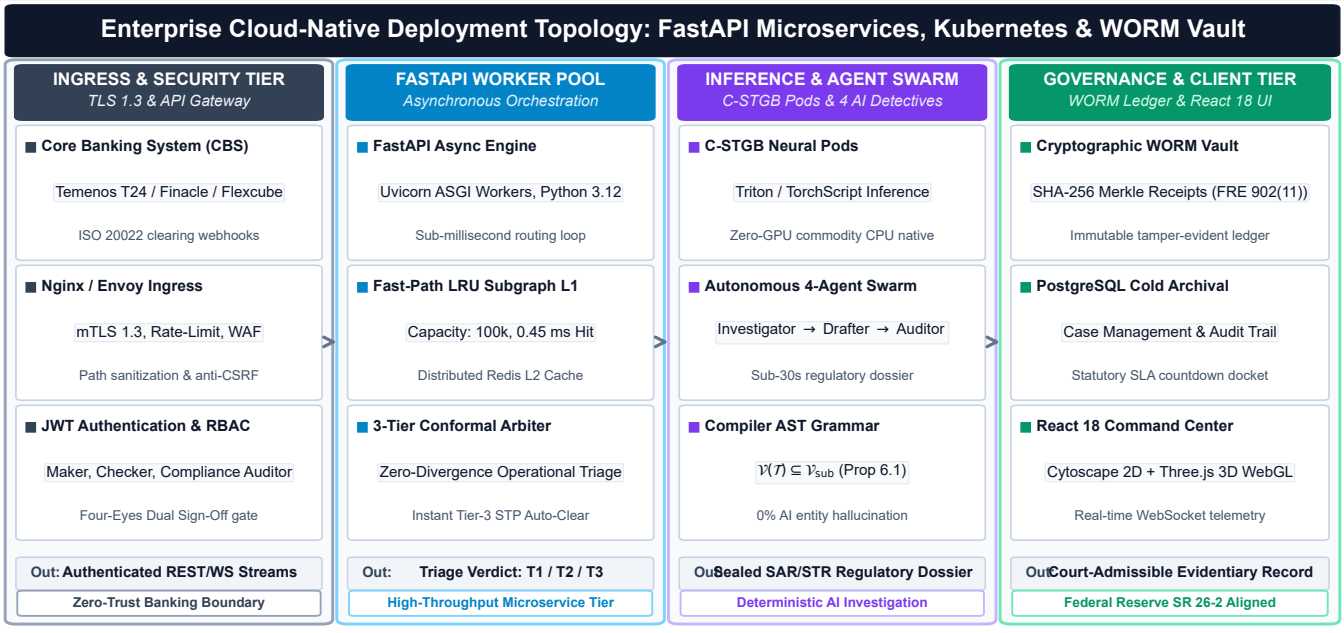

Enterprise Cloud-Native Deployment Topology: FastAPI Microservices, Kubernetes & WORM Vault
INGRESS & SECURITY TIER
TLS 1.3 & API Gateway
Core Banking System (CBS)
Temenos T24 / Finacle / Flexcube
ISO 20022 clearing webhooks
Nginx / Envoy Ingress
mTLS 1.3, Rate-Limit, WAF
Path sanitization & anti-CSRF
JWT Authentication & RBAC
Maker, Checker, Compliance Auditor
Four-Eyes Dual Sign-Off gate
Out: Authenticated REST/WS Streams
Zero-Trust Banking Boundary
FASTAPI WORKER POOL
Asynchronous Orchestration
FastAPI Async Engine
Uvicorn ASGI Workers, Python 3.12
Sub-millisecond routing loop
Fast-Path LRU Subgraph L1
Capacity: 100k, 0.45 ms Hit
Distributed Redis L2 Cache
3-Tier Conformal Arbiter
Zero-Divergence Operational Triage
Instant Tier-3 STP Auto-Clear
Out: Triage Verdict: T1 / T2 / T3
High-Throughput Microservice Tier
INFERENCE & AGENT SWARM
C-STGB Pods & 4 AI Detectives
C-STGB Neural Pods
Triton / TorchScript Inference
Zero-GPU commodity CPU native
Autonomous 4-Agent Swarm
Investigator → Drafter → Auditor
Sub-30s regulatory dossier
Compiler AST Grammar
$\mathcal{V}(\mathcal{T}) \subseteq \mathcal{V}_{\text{sub}}$ (Prop 6.1)
0% AI entity hallucination
Out: Sealed SAR/STR Regulatory Dossier
Deterministic AI Investigation
GOVERNANCE & CLIENT TIER
WORM Ledger & React 18 UI
Cryptographic WORM Vault
SHA-256 Merkle Receipts (FRE 902(11))
Immutable tamper-evident ledger
PostgreSQL Cold Archival
Case Management & Audit Trail
Statutory SLA countdown docket
React 18 Command Center
Cytoscape 2D + Three.js 3D WebGL
Real-time WebSocket telemetry
Out: Court-Admissible Evidentiary Record
Federal Reserve SR 26-2 Aligned